

# Karthikeyan Devadoss

AI Engineer (knowledge & retrieval architecture) | Backend foundation

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## SUMMARY

Applying for AI Engineer, Knowledge Infrastructure at Adyen. Engineer focused on AI solution architecture, with about ten years of backend development since December 2016. Since early 2026, one of a handful of engineers in NRG Energy's initial core AI team, designing and building AI solutions hands-on alongside senior backend work, including production AI triage with a human approval step, adopted by several teams. 4+ years at NRG combining hands-on Java/Spring development with architecture discussions and design decisions alongside the lead. Independently designed and built a public agentic software-delivery platform: approval-gated agents, a deterministic vs LLM decision split, RAG with labelled retrieval evals and MCP tool boundaries.

## CORE SKILLS

|                 |                                                                                                                                                                           |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI stack        | Azure OpenAI, vector retrieval (production, NRG) · Python, Anthropic API, embeddings, Postgres, Playwright (personal project)                                             |
| AI architecture | AI solution design · human-approval & capability boundaries · deterministic vs LLM decision split · RAG & retrieval evaluation (recall/MRR) · MCP & agent tool boundaries |
| Backend         | Java, Spring Boot, REST, GraphQL, JWT/FusionAuth, AWS Lambda/SAM, SQS, Oracle, Docker, Jenkins, JUnit, Kafka, FHIR, Apache Camel                                          |

## EXPERIENCE

Senior Backend Developer — NRG Energy

End client; employed via Remote GmbH

Mar 2022 – present

- Initial core AI team (since early 2026):** one of a handful of engineers in the team, designing and building AI solutions hands-on alongside senior backend work.
- Production AI triage:** Azure OpenAI with vector retrieval and a human email-approval boundary before any action is taken; adopted by several teams; ran informal teaching sessions.
- Architecture:** took part in architecture discussions and design decisions with the lead.
- Hands-on Java/Spring Boot on a large retail-energy platform: REST, GraphQL, JWT/FusionAuth security; 2025 serverless work on AWS Lambda with SAM; SQS with dead-letter queues.
- Mentored offshore developers (sprint breakdown, guidance, PR review); monthly production on-call.

Java Developer Consultant — Blue Cross Blue Shield Association (BCBSA)

Jun 2019 – Mar 2022

- Java/Spring Boot REST services for healthcare integration; hands-on FHIR in member/coverage/claims interoperability.
- Application-side Kafka producers and consumers for domain events; careful handling of sensitive healthcare data.

Java Backend Developer — Marsh (Digital Broker / Bluestream)

Via Innovecture LLC

Dec 2016 – Jun 2019

- Java/Spring APIs and carrier integration (Apache Camel routes, ACORD-oriented mapping), implemented within the technical architect's design.

## PERSONAL PROJECT

Agentic Software Delivery Platform

PERSONAL PROJECT

Independently designed and built

Live demo · GitHub repo

- Deterministic vs LLM split:** deterministic checks kept separate from LLM proposals; a Python agent (Anthropic API) plans and applies changes to a real Spring Boot target via tool calling.
- Human-approval boundary:** structural approval gate; the model has no code path to approve its own change, and approval is bound by content hash to the exact change.
- RAG eval design:** local retrieval (embeddings, cosine similarity, content-hash indexing) with a labelled eval (recall/MRR).
- MCP / tool boundaries:** MCP server for read-only repository tools; independent QA subagent without write tools; durable Postgres event ledger with versioned pricing.

## EDUCATION

B.E. Computer Science — Madras University, Chennai

## LANGUAGES

English — fluent · German — A1-A2